

VotaSJ — Digital Participatory Budgeting · Business Model Canvas · PD1 — INE5458 UFSC 2026/1

1 KEY PARTNERS - São José City Hall — anchor customer & institutional partner - UFSC / Blockchain Lab — academic audit & zk-research - gov.br / Serpro — citizen identity provider (critical dependency) - Polygon Labs — L2 infrastructure & co-marketing - Chainlink — oracle automation (cycle closing) - OpenZeppelin / Hacken — external smart-contract audits - OAB-SC & Public Prosecutor — legal validation - Neighborhood associations — grassroots mobilization - Local media (NDTV, Notícias do Dia) — civic awareness	2 KEY ACTIVITIES - Develop & maintain smart contracts (registry, cycle, tally) - Operate each cycle end-to-end (vote, tally, milestones) — delivers instant-tally VP to City Hall - Recurring external security audits on every major release — manufactures fraud-resistance VP - Maintain gov.br / Serpro identity integration - Train City Hall staff and community agents - Civic communication & mobilization (press, associations) - LGPD & FOI compliance — continuous, not one-off 3 KEY RESOURCES Human 2 Solidity devs, 2 full-stack, 1 UX, 1 security/audit specialist - B2G account executive (long public-sector sales cycle) Intellectual - Audited smart contracts — core asset; without clean audit, no product - VotaSJ protocol (state machine + zk circuits from PD2) Infrastructure - Dedicated Polygon PoS node, IPFS pinning, The Graph subgraph indexer Intangible - Institutional partnership (City Hall MoU, DPO, LGPD DPA) - Brand & public trust — governance product = trust product	4 VALUE PROPOSITIONS Risk reduction - Fraud resistance — immutable on-chain tally; City Hall no longer controls the count - Radical transparency — real-time audit by any citizen, journalist, or councilor - LGPD-compliant privacy — no personal data on-chain; eligibility proven via hash Cost reduction - Replaces BRL 300k–500k/cycle in physical assemblies, poll workers & manual tallying Accessibility - Remote participation — vote from any smartphone, any time during the window - Inclusion via assisted kiosks at CRAS, UPAs, and municipal libraries Performance - Instant tally — result in minutes vs. ~3 weeks of manual reconciliation New offering - Participation SBT — non-transferable proof of civic engagement per cycle Cross-block delivery - Fraud resistance: audited contracts (Key Resource) + external audits (Key Activity) - Kiosks & PWA Channels designed one-to-one for each citizen CS sub-segment - Cost-reduction VP replaces City Hall's BRL 300–500k/cycle — basis of Revenue Streams	5 CUSTOMER RELATIONSHIPS - Self-service (default) — registration, voting & result consultation via PWA - Personal assistance — assisted kiosks operated by municipal staff for digital-inclusion segments - Automated execution tracking — milestone notifications to voters after cycle closes - Community — public proposal commenting scoped by administrative region - Co-creation — proposals originate from citizens; City Hall commits budget, not the idea - Dedicated B2G account management — named technical contact for City Hall renewal - Mix reflects CS: self-service scales to 180k citizens at zero marginal cost - Dedicated B2G retains City Hall — the only CS segment generating Revenue Streams 6 CHANNELS Own - VotaSJ mobile PWA — primary voting channel for citizens - Web portal (são-jose.sc.gov.br integration) — audit, results, press - Assisted physical kiosks (CRAS, UPAs, libraries) — unbanked / non-smartphone citizens - The Graph subgraph + REST API — machine-to-machine for TCE-SC / prosecutors Partner - gov.br OIDC — identity & authentication channel - Neighborhood associations — mobilization & last-mile awareness - Local press (NDTV, Notícias do Dia) — civic communication	7 CUSTOMER SEGMENTS Multi-sided platform Primary (non-paying) ~180,000 eligible citizens of São José/SC (16+) - Job: vote digitally, see result was counted Paying (B2G) - São José City Hall — Planning Sec. + Ombudsman - Job: run a cheap, legitimate, fraud-proof cycle Grassroots - Neighborhood associations (Forquilhinhas, Kobrasol, Barreiros, Campinas) - Job: channel neighborhood demands into a binding process Institutional auditors - TCE-SC and Public Prosecutor's Office - Job: audit in real time with cryptographic certainty Expansion (future) - City halls of Palhoça, Biguaçu, Florianópolis
8 COST STRUCTURE Model: value-driven - Compete on trust & legitimacy (VP), not price — City Hall (CS) buys legitimacy, not software Fixed ~BRL 85k/month - Team payroll (6 people): BRL 65k - Cloud + Polygon node + IPFS: BRL 4k - Legal / LGPD / DPO: BRL 6k - Civic marketing & ops: BRL 10k Variable (per cycle) - Polygon gas: <BRL 0.50/vote (~50k voters) - External security audit: BRL 40k–80k per release — Key Activity manufacturing fraud-resistance VP B2G acquisition ~BRL 30k per city hall (6–12 month sales cycle) Economies of scope - Each new city hall (CS expansion) amortizes audit & dev — scope economies tied to CS growth			9 REVENUE STREAMS Recurring - Annual SaaS subscription — City Hall: BRL 180k–250k/yr (primary revenue) - White-label license — OAB-SC, CREA-SC, unions (recurring) Transactional - Per-cycle setup fee: BRL 40k per cycle executed - Implementation consulting for additional city halls: BRL 60k–120k Seed (non-recurring) - Grants: BNDES, IDB, LabCidades civic-innovation fund Citizens never pay - Monetizing votes would collapse the value proposition CS → Revenue link - City Hall (CS) pays BRL 180–250k/yr because VPs cut their cycle cost from BRL 300–500k — positive ROI	

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ROW 1 Co nceptual Framing	1 PROBLEM • Low turnout (<2% of eligible voters) — requires night-time in-person assemblies • Distrust in the tally — City Hall tallies its own process (interested party, no independent audit) • High operational cost — BRL 300k–500k/cycle in physical ballots, poll workers & manual tallying • Post-vote opacity — citizens cannot verify whether winning project was actually executed	2 SOLUTION • Permissionless-read, permissioned-write public blockchain app on Polygon PoS • Eligibility proof (gov.br + residence) validated off-chain → committed on-chain as credential hash • Proposals, votes & tally run as smart-contract transactions — tamper-evident, publicly auditable • Budget execution milestones recorded on-chain — closes the accountability loop	3 ENTITIES • ~180,000 eligible citizens (São José/SC) — many, heterogeneous, anonymous • City Hall (Planning Sec.) — single operator, interested party • UFSC / Blockchain Lab — academic auditor • TCE-SC (State Court of Audit) — institutional auditor • Public Prosecutor's Office — legal auditor • Polygon validator set (~100) — external, open public network	4 DIVERGENCE • Citizens do not trust City Hall to tally honestly or honor winning proposals • City Hall does not trust citizenry: ballot stuffing, identity fraud, vote buying risks • No neutral third party with legitimacy AND technical capacity to run the process • Every cycle ends with opposition fraud accusations — unprovable either way • → Blockchain fits: trust problem is horizontal & political	5 MOTIVATION Gains • Citizen: digital participation + verifiable vote + SBT + execution follow-up • City Hall: lower costs, higher turnout, defense against fraud claims • Auditors: real-time cryptographic trail instead of paper reconstruction Attack deterrence • Sybil: blocked by gov.br CPF → 1 credential hash → 1 vote • Admin abuse: 3-of-5 multisig + 14-day upgrade timelock • Vote buying: zk-proof of eligibility removes 'prove who you voted for' ability • Censorship: public Polygon L2 — City Hall cannot unilaterally take offline
	6 NETWORK PEERS • Eligible citizen (wallet + gov.br credential) — submit proposal, cast 1 vote/cycle, read all • City Hall multisig (3-of-5 with UFSC + TCE) — open cycle, set budget/deadlines, record milestones • Auditor (UFSC / TCE-SC multisig signer) — co-sign admin ops, technical veto • Public observer (any wallet) — read-only, no write rights • Polygon validator — external; provides consensus & finality, not a VotaSJ role	7 TRANSACTIONS • registerVoter(addr, credentialHash) — onboard eligible citizen • revokeVoter(addr, reason) — remove eligibility (death, fraud, loss of residence) • openCycle(opensAt, closesAt, budgetWei) — start cycle (multisig) • submitProposal(ipfsCid) — citizen submits proposal; metadata pinned to IPFS • vote(proposalId) — 1 vote per citizen per cycle • closeCycle() — deterministic tally, winner emitted via event • recordMilestone(...) (PD3) — on-chain execution progress commitment • Volume: ~50k tx/cycle; 1 cycle/year initially	8 DATA • Credential hash (keccak256) — on-chain, low volume, HIGH criticality • Proposal IPFS CID + submitter — on-chain, low volume, HIGH criticality • Proposal metadata (title, description, budget, images) — IPFS only, medium volume • Vote (voter addr, proposal id, cycle id) — on-chain, HIGH volume (~50k/cycle), CRITICAL • Tally result (winner, total votes) — on-chain, low volume, CRITICAL • Execution milestones — on-chain, low volume, HIGH criticality • Personal data (CPF, name, address) — NEVER on-chain (LGPD)	9 TYPE OF PROCESSING Distributed calculation • Voter eligibility check, one-vote enforcement, tally computation • Cycle state machine: Pending → Open → Closed Distributed storage (event log) • Immutable audit trail of every registration, vote & cycle closure Oracles • gov.br — off-chain identity → on-chain via City Hall multisig relayer • Chainlink Automation — triggers closeCycle() at closesAt without human action	10 VALUE No fungible / speculative token Civic & institutional value • Credential SBT (ERC-5192) — non-transferable proof of municipal citizenship • Participation badge SBT (per cycle) — permanent non-transferable civic engagement record • Budget commitment (wei) — symbolic of BRL value; actual funds move off-chain via SIG Explicitly ruled out • Fungible vote-token — monetizing votes invalidates the entire value proposition
	11 NETWORK DYNAMICS Consensus (inherited) • Polygon PoS — HotStuff BFT, ~100 public validators, ~2s finality Application-layer rules (smart contract) • registerVoter: only multisig admin, non-zero addr, not already registered • submitProposal: cycle Open, sender registered, CID non-empty • vote: cycle Open, within [opensAt, closesAt], sender registered & not yet voted • openCycle: opensAt < closesAt > now, previous cycle Closed • closeCycle: only after closesAt, max(votes) computed deterministically Upgrade rule • UUPS proxy + 14-day timelock — no upgrades during active cycle Admin rule • 3-of-5 multisig (City Hall ×3, UFSC ×1, TCE-SC ×1) — no single signer can act alone		12 POINTS TO VERIFY (off-chain) • Legal feasibility — OAB-SC & municipal legal dept must confirm digital process satisfies Organic Law • gov.br integration — exact Serpro API flow, rate limits & SLA need validation • LGPD opinion — municipal DPO must confirm CPF-derived hash is not personal data processing • zk-proof of eligibility (PD2) — Semaphore/zk-SNARK circuit needs prototyping on Polygon Amoy • Real turnout — >5% target is hypothesis; needs pilot validation (PD3) • Assisted kiosk feasibility — staff, location & device custody must be negotiated with City Hall • Gas-cost budget — ~430k gas/citizen-flow measured on Hardhat; must re-measure on Amoy		